GÖRKEM AKGÜL

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EDUCATION

BERNSTEIN CENTER FOR COMPUTATIONAL NEUROSCIENCE BERLIN

Master of Science in Computational Neuroscience (*2024–2027*) **German Grade:** 1.7

Joint degree program offered by Humboldt-Universität zu Berlin, Charité Universitätsmedizin Berlin and Technische Universität Berlin, organized by the BCCN Berlin.
Advanced Coursework: Stochastic Processes in Neuroscience, Stochastic Modelling in Neuroscience, Models of Biological Neural Networks.

USKUDAR UNIVERSITY | *Double Major*

Bachelor of Science in Software Engineering (*2020–2024*) **GPA:** 3.83/4.0

Thesis Title: Segmentation of Brain Tumor from MRI Images with Deep Convolutional Neural Networks – Supervisor: Kristin Benli

Bachelor of Science in Molecular Biology and Genetics (*2019–2023*) **GPA:** 3.79/4.0

Thesis Title: Design of Neuron-Specific Lentiviral Vectors and Investigation of Gene Therapy Applications in Spinal Muscular Atrophy – Supervisor: Cihan Tastan – Thesis Poster: DOI.

HONORS AND AWARDS

International Scholarship Programme | *Avicenna Studienwerk* *Oct 2024 – Jan 2027*

- Selected as one of 16 recipients for the scholarship funded by the Federal Foreign Office of Germany, including **a monthly stipend of €1,044** to enable my Master's studies in Computational Neuroscience.

Department Ranking Award | *Uskudar University* *June 2023*

- Ranked 1st in the graduating class of the Department of Molecular Biology and Genetics.

Erasmus+ Student Mobility Grant | *European Commission* *June 2022*

- Awarded a **€1,500** Erasmus+ Student Mobility Grant for a research internship at MPI CBS.

RESEARCH EXPERIENCE

BCCN Research Rotations

Max Planck Institute for Human Cognitive and Brain Sciences *Aug 2026 – Present*

Title: Geometric Exploration of Learned Internal Representations in Vision Systems

- Currently extending prior work by applying geometric analysis to study representational alignment between fMRI data and vision models (e.g., CLIP, SLIP, DINO).

Supervisor: Nico Scherf

Kempner Institute at Harvard University *Jan 2026 – Aug 2026*

Title: Mechanistic Interpretability and Representation Alignment in Autoregressive Speech-Language Models

- Implemented mechanistic interpretability methods to probe syntactic and lexical circuitry in autoregressive speech-language models introduced by Tuckute et al., 2025, using the ZeroSpeech dataset.
- Analyzed representation alignment using RSA, CKA, and voxel-wise encoding in the human auditory cortex. This work forms the basis of my Master's thesis.

Supervisor: Greta Tuckute

Technical University of Berlin *Oct 2025 – Jan 2026*

Title: Interpretability Analysis of Bilingual Brain-Informed Fine-Tuning in BERT

- Investigated the mechanistic and representational effects of brain-informed fine-tuning in English and Chinese BERT models, extending Negi et al. (2025) using layer-wise RSA, logit-lens analysis, causal tracing, path patching, and structural probes.

Supervisor: Fatma Deniz

MPI for Human Cognitive and Brain Sciences | *Neural Data Science and Statistical Computing Group*

Research Assistant *Apr 2025 – Present*

- Project BIOWIN: Analyzed high-dimensional spectroscopic measurements analogous to fMRI data, clustering latent representations using classical methods, autoencoders, and variational autoencoders.

Guest Researcher *Oct 2022 – Sep 2024*

- Continued collaborating remotely with the group, extending image-segmentation work on neuronal fluorescence images.
- Investigated functional and structural connectivity patterns in Human Connectome Project data using graph neural networks as part of my BSc Software Engineering graduation project.

Erasmus+ Research Intern *Jul 2022 – Oct 2022*

- Implemented and compared vision transformers and convolutional neural networks, evaluating their performance and exploring techniques to enhance their effectiveness on biological segmentation tasks.

Supervisor: Nico Scherf

University College London | *Centre for Medical Image Computing*

Summer Research Intern *Jul 2023 – Nov 2023*

- Developed tree-based ensemble models for Alzheimer's Disease (AD) using multimodal biomarkers, including MRI, PET, DTI, genetics, and cognitive tests, to predict key AD outcomes.
- Evaluated predictor importance to identify key biomarkers associated with AD progression in clinical trials.

Supervisors: Neil Oxtoby and Maitrei Kohli

Uskudar University | *Transgenic Cell Technologies and Epigenetics Application and Research Center*

Guest Researcher *Jan 2021 – Jun 2023*

- Played a significant role in the preparation of grant proposals for TUBITAK and TUSEB. This effort resulted in the full funding of the project titled **"In Vitro and Ex Vivo Study of New Generation CRISPR-Prime Editing that targets Exon 7/Intron 7 of SMN2 Gene Regulation in Spinal Muscular Atrophy (SMA) and Approaches in SMN1 Encoding Motor Neuron Cell Specific Neural Lentivirus"** by TUSEB.
- Contributed to CRISPR-based and XNA-based therapeutic approaches for Spinal Muscular Atrophy (SMA) and Retinitis Pigmentosa.

Supervisor: Cihan Tastan

PATENTS

1. CRISPR-PE System for Retinol Dehydrogenase 12 (Rdh12) Gene Mutations for Use in the Treatment of Retinitis Pigmentosa (RP) Disease *(pending, No: TR2021022284A2)*
2. Xeno Nucleic Acid Antisense-Oligonucleotide (XNA-ASO) Sequences for the Genetic Treatment of USH2A Induced Retinitis Pigmentosa Disease *(pending, No: TR2022001648A2)*
3. Neural Lentiviral Vector Containing the Functional Abca4 Gene for Use in the Treatment of Retinitis Pigmentosa (RP) Disease *(pending, No: TR2022000448A2)*
4. Neurotrophic Factor and Optogenetic System-Based Lentiviral System for Use in Retinal Diseases *(pending, No: TR2022021763A2)*
5. Methods for the Treatment of Spinal Muscular Atrophy *(pending, No: WO2025144141A1)*